L51 – MOOD JOURNAL

OS Linux Lab Mini-Project Report | Project ID: L51

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Division	H	**Course**	OS Linux Lab 260120510
Project	Mood Journal	**Platform**	Ubuntu on WSL 2

1. Project Overview

The Mood Journal is a Bash shell-scripting project that allows a user to record a daily journal entry with a mood score from 1 to 5. It also provides weekly mood statistics, keyword search, word counting, data viewing, backup creation and activity logging.

Objectives

• Record daily mood scores and journal entries in date-based text files.

• Calculate weekly mood averages and identify best and worst mood days.

• Search saved entries, count words, and view stored journal data.

• Create compressed backups and schedule a daily reminder using cron.

2. Main Menu

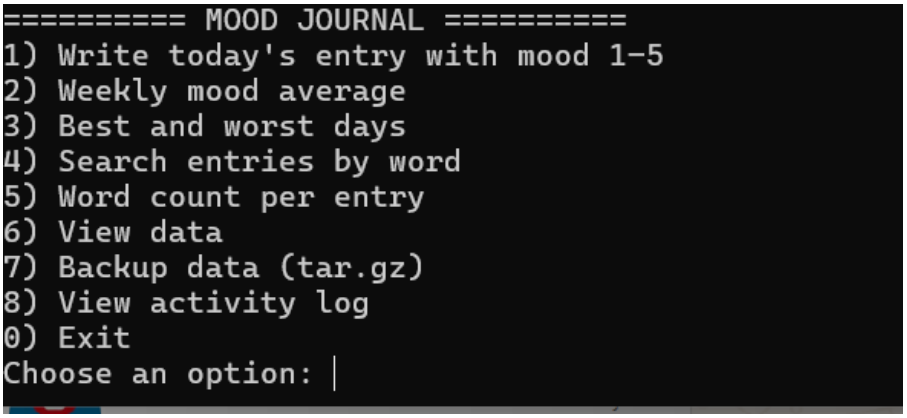
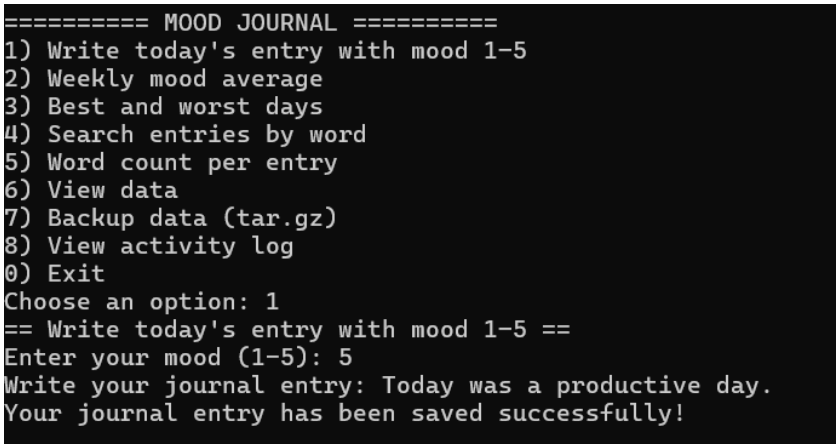


Figure 1. Main menu of the Mood Journal application.

3. Implementation and Feature Testing

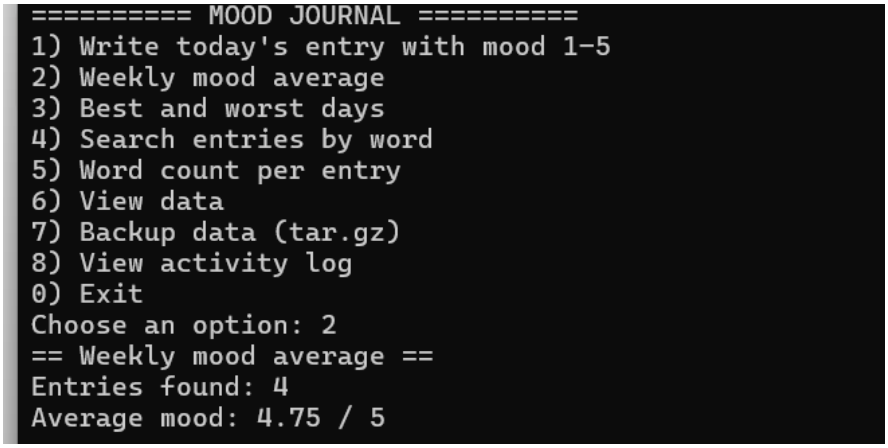
The following screenshots show the implemented journal functions and their terminal outputs.

Feature 1: Saving a journal entry with mood score 5



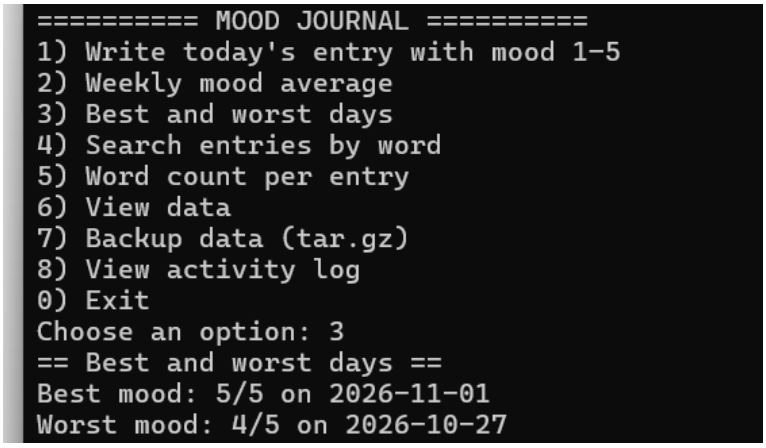
Terminal output captured during project testing.

Feature 2: Weekly mood average calculation



Terminal output captured during project testing.

Feature 3: Best and worst mood days



Terminal output captured during project testing.

4. Search, Word Count and Journal Data

Feature 4: Searching journal entries by keyword

```
========== MOOD JOURNAL ==========
1) Write today's entry with mood 1-5
2) Weekly mood average
3) Best and worst days
4) Search entries by word
5) Word count per entry
6) View data
7) Backup data (tar.gz)
8) View activity log
0) Exit
Choose an option: 4
== Search entries by word ==
Enter word to search: productive
3:Entry: Today was productive and good.
File: 2026-10-10.txt
3:Entry: Today was productive and good.
File: 2026-10-15.txt
3:Entry: Today was productive and good.
File: 2026-10-20.txt
3:Entry: Today was productive and good.
File: 2026-10-25.txt
7:Entry: Today was a productive day.
File: 2026-11-01.txt
```

Terminal output captured during project testing.

Feature 5: Word count for journal entries

```
========== MOOD JOURNAL ==========
1) Write today's entry with mood 1-5
2) Weekly mood average
3) Best and worst days
4) Search entries by word
5) Word count per entry
6) View data
7) Backup data (tar.gz)
8) View activity log
0) Exit
Choose an option: 5
== Word count per entry ==

Date: 2026-10-08
Entry: Today was difficult and stressful.
Word count: 5

Date: 2026-10-09
Entry: Today was excellent and very enjoyable.
Word count: 6

Date: 2026-10-10
Entry: Today was productive and good.
Word count: 5
```

Terminal output captured during project testing.

Option 6: Viewing saved journal data

```
========== MOOD JOURNAL ==========
1) Write today's entry with mood 1-5
2) Weekly mood average
3) Best and worst days
4) Search entries by word
5) Word count per entry
6) View data
7) Backup data (tar.gz)
8) View activity log
0) Exit
Choose an option: 6

========== 2026-10-08.txt ==========
     1  Date: 2026-10-08
     2  Mood: 1
     3  Entry: Today was difficult and stressful.
     4  --------------------------------

========== 2026-10-09.txt ==========
     1  Date: 2026-10-09
     2  Mood: 5
     3  Entry: Today was excellent and very enjoyable.
     4  --------------------------------

========== 2026-10-10.txt ==========
     1  Date: 2026-10-10
     2  Mood: 4
```

Terminal output captured during project testing.

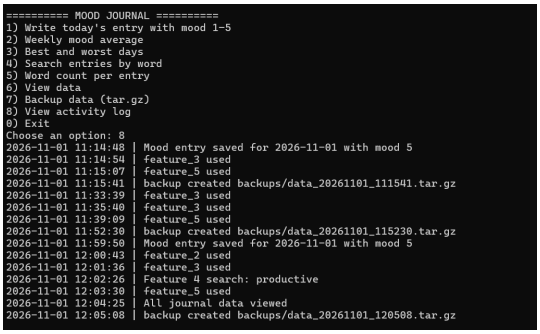
5. Backup, Activity Log and Scheduling

Option 7: Creating a compressed TAR.GZ backup

```
========== MOOD JOURNAL ==========
1) Write today's entry with mood 1-5
2) Weekly mood average
3) Best and worst days
4) Search entries by word
5) Word count per entry
6) View data
7) Backup data (tar.gz)
8) View activity log
0) Exit
Choose an option: 7
Backup saved: backups/data_20261101_120508.tar.gz (4.0K)
```

Terminal output captured during project testing.

Option 8: Activity log showing recorded operations



```
========== MOOD JOURNAL ==========
1) Write today's entry with mood 1-5
2) Weekly mood average
3) Best and worst days
4) Search entries by word
5) Word count per entry
6) View data
7) Backup data (tar.gz)
8) View activity log
0) Exit
Choose an option: 8
2026-11-01 11:14:48 | Mood entry saved for 2026-11-01 with mood 5
2026-11-01 11:14:54 | feature_3 used
2026-11-01 11:15:07 | feature_5 used
2026-11-01 11:15:41 | backup created backups/data_20261101_111541.tar.gz
2026-11-01 11:33:39 | feature_3 used
2026-11-01 11:35:40 | feature_3 used
2026-11-01 11:39:09 | feature_5 used
2026-11-01 11:52:30 | backup created backups/data_20261101_115230.tar.gz
2026-11-01 11:59:50 | Mood entry saved for 2026-11-01 with mood 5
2026-11-01 12:00:43 | feature_2 used
2026-11-01 12:01:36 | feature_3 used
2026-11-01 12:02:26 | Feature 4 search: productive
2026-11-01 12:03:30 | feature_5 used
2026-11-01 12:04:25 | All journal data viewed
2026-11-01 12:05:08 | backup created backups/data_20261101_120508.tar.gz
```

Terminal output captured during project testing.

Cron Configuration

A crontab entry is configured to run the project script every day at 9:00 PM. The script appends a daily journal reminder and generates a weekly mood report on Sunday. The cron service was checked and reported as active (running).

```
0 21 * * * /bin/bash $HOME/linux_project/moodjournal/main.sh --cron >>
$HOME/linux_project/moodjournal/logs/cron.log 2>&1
```

Commands and Learning Outcomes

Commands used include date, awk, grep, wc, ls, sed, tar and mkdir. Through this project, the student practised Bash scripting, functions, loops, conditional statements, input validation, file handling, logging, compressed backups and cron scheduling.

Conclusion

The Mood Journal menu and its five main features were implemented and tested. Journal data viewing, backup creation and activity logging were also demonstrated. The project archive was created as E11_L51_Himani_Gohel.tar.gz.

Project Guide: Dr. Anjum Y. Mansuri, Assistant Professor, Computer Applications, LJ Institute of Computer Applications, LJ University, Ahmedabad.